

CARPENTRAS, France

WMO: 075860

Lat: 44.0829N

Lon: 5.0592E

Elev: 105

StdP: 100.07

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-3.6	-11.0	1.5	-1.6	-8.3	1.9	0.2	9.7	6.5	8.8	7.4	1.3	110	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	36.1	21.2	34.4	20.9	32.8	20.3	22.7	32.7	21.9	31.5	21.2	30.4	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.6	25.8	18.9	13.9	25.2	18.1	13.2	24.4	67.3	32.8	64.3	31.4	61.6	30.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.4	6.5	DB	-7.5	38.7	2.4	1.6	-9.2	39.9	-10.6	40.8	-12.0	41.7	-13.7	42.9	
			WB	-8.0	24.4	2.4	0.8	-9.8	24.9	-11.2	25.4	-12.6	25.8	-14.3	26.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	5.7	6.6	10.3	13.7	17.7	22.0	24.4	23.7	19.7	15.3	9.8	6.1
	DBStd	7.35	3.51	3.55	3.15	2.85	2.92	2.94	2.56	2.68	2.78	3.40	3.77	3.66
	HDD10.0	460	139	102	33	4	0	0	0	0	4	48	129	
	HDD18.3	1912	392	329	248	140	47	3	1	1	17	102	255	378
	CDD10.0	2152	6	6	44	116	237	361	446	424	291	169	43	8
	CDD18.3	563	0	0	0	2	26	115	188	166	57	9	0	0
	CDH23.3	6776	0	0	4	66	379	1426	2281	1916	628	75	0	0
	CDH26.7	3107	0	0	0	8	98	649	1176	962	205	7	0	0
(14) Wind	WSAvg	2.4	2.3	2.6	2.6	2.7	2.6	2.5	2.6	2.4	2.2	2.1	2.2	2.1
(15) Precipitation	PrecAvg	737	61	40	44	60	59	43	35	44	93	99	107	53
	PrecMax	1112	175	112	106	123	145	100	96	117	262	176	260	174
	PrecMin	503	6	0	0	15	7	5	2	1	18	17	19	0
	PrecStd	173	44	30	30	35	36	26	26	27	51	45	70	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	19.3	23.8	27.8	31.9	36.9	37.6	38.5	33.4	27.6	20.7	17.0
		MCWB	11.6	11.5	13.0	16.4	19.3	21.8	22.0	21.9	20.4	18.1	14.9	12.4
	2%	DB	15.0	16.6	21.0	25.2	29.3	34.5	35.9	36.2	31.0	25.4	18.8	15.2
		MCWB	10.5	10.4	12.6	15.1	18.2	21.1	21.3	21.1	19.4	18.0	14.0	11.4
	5%	DB	13.4	14.8	19.2	23.3	27.3	32.6	34.5	34.0	29.1	23.6	17.5	13.8
		MCWB	9.6	9.5	11.8	14.5	17.4	20.4	20.8	20.9	18.7	17.1	13.4	10.4
	10%	DB	11.9	13.1	17.4	21.2	25.4	30.6	32.9	32.1	27.2	22.0	16.1	12.3
		MCWB	8.5	8.5	10.9	13.2	16.5	19.5	20.2	20.3	18.2	16.1	12.4	9.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	12.2	14.4	17.4	20.6	23.8	23.9	23.4	21.8	19.8	16.1	13.2
		MCDB	15.9	17.3	20.3	25.4	29.0	35.2	35.0	33.9	29.1	25.3	19.0	16.1
	2%	WB	11.1	11.1	13.3	16.0	19.2	22.1	22.7	22.5	20.8	18.7	14.8	12.0
		MCDB	14.1	15.4	19.5	24.0	27.1	31.9	32.9	32.3	27.8	23.7	17.8	14.5
	5%	WB	10.0	10.1	12.4	14.9	18.2	21.1	21.9	21.7	20.0	17.7	13.9	11.0
		MCDB	12.9	13.8	18.0	21.9	25.5	30.5	31.8	31.2	26.8	22.3	16.8	13.4
	10%	WB	8.8	9.0	11.5	13.8	17.1	20.1	21.1	20.9	19.1	16.7	13.0	9.7
		MCDB	11.5	12.4	16.5	19.9	23.9	28.9	30.4	30.3	25.3	20.9	15.5	11.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.2	12.7	13.0	13.3	14.6	15.3	15.3	13.6	11.6	9.9	9.5
		MCDBR	11.1	14.2	16.0	17.2	17.1	18.3	18.4	19.2	17.0	13.7	11.3	10.3
	5% WB	MCWBR	7.2	8.7	8.8	8.7	7.7	7.2	6.8	7.2	7.1	7.1	7.1	6.8
		MCWBR	10.2	12.7	14.1	15.8	15.3	17.1	16.9	17.1	14.4	12.2	9.9	9.0
(40) Clear-Sky Solar Irradiance	taub		0.313	0.332	0.370	0.391	0.401	0.400	0.393	0.390	0.379	0.377	0.340	0.311
		taud	2.473	2.417	2.333	2.295	2.316	2.343	2.363	2.376	2.397	2.403	2.455	2.490
	Ebn at Noon		813	857	863	872	872	871	873	863	844	792	769	780
		Edh at Noon	71	89	111	124	126	123	119	114	103	90	73	64
(44) All-Sky Solar Radiation	RadAvg		1.68	2.63	3.90	5.06	6.07	7.08	7.10	6.10	4.64	2.93	1.84	1.45
	RadStd		0.18	0.27	0.35	0.42	0.47	0.34	0.36	0.23	0.29	0.28	0.23	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page